Physics 4730, Computational Physics: Statistics and Data Science (Fall 2026)

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Description This advanced course is focused on development of a skillset suitable for the analysis and statistical interpretation of scientific data. Students may write programs in both general-purpose (Python, C++ . . .) and symbolic manipulation languages (SymPy, Mathematica . . .). Specific topics include data mining, basic probability distributions (binomial, Poisson, Gaussian), error propagation, parameter estimation by χ^2 minimization and maximum likelihood, comparison of distributions, Bayesian inference, machine learning and neural networks. Current "hot topics" may be presented at the survey level. Exercises will be chosen to illustrate methods applicable to active research topics. This is a required course for undergraduate students working towards the Computational Physics Emphasis.

Prerequisites PHYS 3010 AND (MATH 2250 OR (MATH 2270 AND MATH 2280))

Lectures/Labs TTh 08:35 – 10:30
LSSB W3262

Web Page **https://jwbelz.github.io/phys4730/**

Text There is no textbook for the course; All supplementary information can be obtained online.

Grading Grading for the course will break down as follows:

In-Class Exercises	25%
Homework	50%
Final Project	25%

Drop/Add Last day to add, August 28.
Last day to drop, September 4.
Last day to withdraw, November 11.

Lecture and Lab Schedule – Physics 4730, Fall 2026

Week 01	08/25	Course Introduction
	08/27	Python and Linux Review
Week 02	09/01	Linear Algebra, Pseudoinverse
	09/03	Random Variables
Week 03	09/08	Basic probability distributions. Binomial, Poisson, Gaussian
	09/10	Error Propagation and Covariance
Week 04	09/15	Parameter Estimation: Method of Moments
	09/17	Parameter Estimation: χ^2 Minimization
Week 05	09/22	Parameter Estimation: Maxmimum Likelihood I
	09/24	Maximum Likelihood II
Week 06	09/29	χ^2 Minimization II: Goodness-of-Fit
	10/01	Fisher Information
Week 07	10/06	Confidence Intervals
	10/08	Empirical Maximum Likelihood I
Week 08	10/13	NO CLASS, FALL BREAK
	10/15	NO CLASS, FALL BREAK
Week 09	10/20	Empirical Maximum Likelihood II
	10/22	Comparing Distributions: KS Test
Week 10	10/27	Comparing 2D RV Distributions
	10/29	Comparing Distributions with Unknown Parameters
Week 11	11/03	Final Project Assignment
	11/05	Bayes' Theorem
Week 12	11/10	Introduction to Machine Learning
	11/12	Decision Trees
Week 13	11/17	Dealing with Real Data, Nearest Neighbor Algorithm
	11/19	Diagnostics and Evaluation of Supervised Learning
Week 14	11/24	Neural Network Basics
	11/26	NO CLASS, THANKSGIVING HOLIDAY
Week 15	12/01	More Neural Networks
	12/03	Neural Networks and Data Analysis
Week 16	12/08	Final project help
	12/10	Final project help